ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

PROJECT MANAGEMENT

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 3. Section A: objective questions (30) · Section B: project concepts and the project life cycle, scope and the WBS, scheduling (Gantt charts and the critical path), budgets and resources, risk, communication and teams (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A project is best described as …

A a temporary effort with a definite start and end that creates a unique result.
B the ongoing daily work of a business.
C any task done by one person.
D a permanent department.

1.1.2 The three constraints of the "project triangle" are …

A people, profit and planet.
B input, process and output.
C risk, quality and tax.
D scope, time and cost.

1.1.3 The longest path of dependent activities through a project network, which determines the shortest project duration, is the …

A float path.
B critical path.
C milestone.
D work package.

1.1.4 A bar chart that shows activities against time is a …

A pie chart.
B histogram.
C flowchart.
D Gantt chart.

1.1.5 A significant event in a project with zero duration, e.g. "system goes live", is a …

A deliverable.
B milestone.
C resource.
D risk.

1.1.6 The hierarchical breakdown of all the work in a project into smaller parts is the …

A SWOT.
B CPM.
C WBS.
D KPI.

1.1.7 Uncontrolled growth of the project's requirements after it has started is called …

A scope creep.
B float.
C crashing.
D slack.

1.1.8 The phase in which the project is formally approved and the project charter is written is …

A planning.
B initiation.
C execution.
D closure.

1.1.9 The amount of time an activity can be delayed without delaying the project is the …

A lead time.
B duration.
C critical time.
D float (slack).

1.1.10 A person or group affected by the project or with an interest in it is a …

A stakeholder.
B supplier only.
C competitor.
D sponsor only.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Activities on the critical path have zero float.

1.2.2 A project sponsor provides the money and support for a project.

1.2.3 Risk management only starts after a problem has occurred.

1.2.4 Lessons learnt should be recorded when a project is closed.

1.2.5 Reducing the scope of a project usually increases its cost.

1.2.6 A contingency fund is money set aside for unexpected costs.

1.2.7 In a Gantt chart, activities that run at the same time are shown on top of one another as one bar.

1.2.8 Good communication with stakeholders reduces misunderstandings.

1.2.9 A SMART objective must be specific, measurable, achievable, relevant and time-bound.

1.2.10 The project manager should do all the technical work alone.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	A product or result that must be handed over to the client	A	RACI matrix
1.3.2	Document that formally authorises a project	B	baseline
1.3.3	An activity that must finish before another can start	C	project charter
1.3.4	People, equipment and money needed for the project	D	change control
1.3.5	An uncertain event that could affect the project	E	crashing
1.3.6	Table that lists who is responsible, accountable, consulted and informed	F	key performance indicator
1.3.7	Adding resources to finish activities faster	G	resources
1.3.8	Measure used to check whether the project meets its goals	H	risk
1.3.9	Formal process for approving changes to the project	I	deliverable
1.3.10	Approved version of the plan used to measure progress	J	predecessor

(10)

TOTAL SECTION A: [30]

SECTION B

Case study: Computer lab for a rural school
A company has donated R250 000 to set up a 20-computer lab at Sizanani Secondary School. Your team of NC(V) students has been asked to manage the project. The lab must be ready before the start of the third term, 10 weeks from now. The work includes preparing the classroom (burglar bars, electrical points), buying computers and furniture, installing a network and software, and training two teachers.

QUESTION 2: PROJECT CONCEPTS AND THE LIFE CYCLE

2.1 Name the FOUR (or five) phases of the project life cycle and describe ONE activity in EACH phase for the computer lab project. (8)

2.2 Write a SMART objective for this project. (3)

2.3 Identify FOUR stakeholders in the project and state the interest of each. (4)

[15]

QUESTION 3: SCOPE AND SCHEDULING

3.1 Draw a work breakdown structure (WBS) for the project with the project at level 0 and at least FOUR level-1 work packages, each with TWO level-2 tasks. (8)

3.2 The table shows some of the activities of the project.

Activity	Description	Duration (days)	Predecessor(s)
A	Plan the lab layout	2	–
B	Electrical work and burglar bars	4	A
C	Order and receive equipment	3	A
D	Install computers and network	6	B, C
E	Test the network and software	2	D
F	Train the teachers	3	D
G	Official hand-over	1	E, F

3.2.1 Draw a network diagram (activity-on-node) for the activities. (6)

3.2.2 List all the paths through the network and their durations. (3)

3.2.3 Identify the critical path and the minimum duration of the project. (2)

3.2.4 Calculate the float of activities C and E. (2)

3.2.5 Draw a Gantt chart for the activities, assuming each activity starts as early as possible. (4)

[25]

QUESTION 4: BUDGET, RESOURCES AND RISK

4.1 The budget items are: 20 computers at R8 500 each, furniture R25 000, network equipment and cabling R18 000, electrical work R12 000, software licences R6 000 and training R4 000. Calculate the total cost and how much of the R250 000 remains as a contingency. (4)

4.2 Identify FOUR risks to the project and give a way to reduce (mitigate) EACH one. (8)

4.3 Explain what a risk register is and name THREE columns it should contain. (3)

[15]

QUESTION 5: COMMUNICATION, TEAMS AND CLOSURE

5.1 Give THREE ways the project team can communicate progress to the school and the donor. (3)

5.2 Name FOUR qualities of a good project manager. (4)

5.3 The principal asks the team to also install a printer room halfway through the project. Explain how the team should handle this request. (4)

5.4 Give FOUR activities that must take place when the project is closed. (4)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER